Omar Khattab

GitHub

Electric Rail Operator | Computer Science Student

Contact

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Technical Skills

Operational Systems

CBTC Train Control
CCTV | Radio Communications
Incident Response

Soft Skills

Multitasking
Analytical Reasoning
Critical Thinking
Communication
Team Leadership

Frameworks

Spring Boot

Developer Tools

Git
Visual Studio
IntelliJ IDEA
Cursor

Libraries

Openpyxl
Selenium
Requests

Profile

Motivated and detail-oriented **Electric Rail Operator** and **Computer Science student** with proven ability to manage complex systems and safety-critical operations under pressure. Experienced in the **Confederation Line CBTC environment**, communication with the **Transit Operations Control Centre (TOCC)**, and real-time problem solving. Known for strong multitasking, balancing full-time rail operations with full-time studies, and an unwavering commitment to safety, precision, and teamwork within OC Transpo.

Professional Experience

Electric Rail Operator (ERO)
OC Transpo – Ottawa, ON April 2025 – Present

- Operated and monitored LRT vehicles under **TOCC direction** using **CBTC train-control systems**, always prioritizing **personal and public safety** in accordance with **ERO Rule 1.1 (Safety is the Priority)**.
- Maintained continuous radio communication with TOCC, including **repeat-backs of movement authority** and confirmation per **ERO Rule 3.12.3 (Receiving and Repeating Communications)**.
- **Reported and documented** operational irregularities, equipment faults, and service disruptions as required under **ERO Rule 1.5 (Reporting)** to ensure transparent, safe, and reliable operations.
- Responded effectively to alarms and system abnormalities, coordinating with TOCC to resolve issues while maintaining passenger safety and service continuity.
- **Executed emergency procedures** by switching to **ATPM mode** and proceeding at restricted speed during an **intruder broadcast,** following **ERO Rule 3.14 (Emergency Communications)**.
- Demonstrated strong **situational awareness, rule compliance, and communication discipline**, consistently upholding OC Transpo's commitment to **safety, reliability, and service excellence**.

Bus Operator
OC Transpo – Ottawa, ON March 2024 – April 2025

- Operated city buses safely and efficiently in accordance with ATU 279 protocols and City o Ottawa procedures.
- Maintained calm, clear communication with Control Centre staff during detours, emergencies, and schedule disruptions.
- Developed strong **customer-service and conflict-resolution** skills under high-demand conditions.
- Gained hands-on understanding of **transit-control communication systems** and **incident-response coordination**.

Education

Honours BSc Computer Science
University of Ottawa 2020 – Present

- Final year in progress — **GPA 7.0+ / 10.0**
- Coursework includes: Operating Systems, Digital Systems, Artificial Intelligence, Database Design, and Software Engineering.

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Some Applied Projects

GymBroChatbot **April 2025**

- **Developed a Python-based AI** fitness & nutrition chatbot using **Gradio** for UI and **Groq API** for intelligent responses**.**
- **Built interactive features** for workout and meal-plan generation; **implemented customizable conversation** settings to adapt to user preferences.
- **Demonstrated full-stack capabilities**: backend logic, front-end interface, API integration and deployment readiness.

Dental Clinic Management System **April 2022**

- **Relational Database Creation:** Established a structured SQL database to catalog and retrieve dental patient data efficiently.
- **Robust Backend Solutions:** Developed a reliable backend using Java, guaranteeing data integrity and timely access.
- **Framework Utilization:** Leveraged the Spring Boot framework for UI design, providing users with a seamless and intuitive clinic management platform.

Car Rental app **December 2021**

- **Database Design & Management:** Engineered a robust Firebase database to store and manage car rental data.
- **Backend Development:** Utilized Java to create a dynamic and responsive backend, ensuring data accuracy and accessibility.
- **Intuitive User Interface:** Employed Android Studio for UI design, ensuring a user-friendly experience for customers.

Professional Qualifications

- Exceptional **situational awareness** and **colour differentiation**, vital for monitoring multiple live control interfaces in high-pressure environments.
- Deep commitment to **OC Transpo's values of safety, reliability, and operational excellence**, consistently demonstrated in day-to-day rail operations.
- Proven **time-management and multitasking abilities**, balancing full-time rail operations with full-time university studies while maintaining strong academic performance.
- Solid **technical foundation** through advanced coursework in computer systems, automation, and data analysis skills directly transferable to control system monitoring and troubleshooting.
- Strong **understanding and adherence to the Electric Rail Operator Rule Book**, ensuring safe and compliant operation in all conditions.
- Recognized for **focus, consistency, and composure** during live service events requiring real-time coordination with TOCC.
- Dedicated to **continuous improvement and professional growth** within Ottawa's expanding rail network and control operations.

References available upon request